

$$\begin{array}{c} a_j \\ \hline d_j \end{array} = \begin{array}{c} (\cdot \star \bar{h}) \downarrow_2 \\ \hline (\cdot \star \bar{g}) \downarrow_2 \end{array} \times \begin{array}{c} a_{j-1} \end{array} \iff \begin{array}{c} a_{j-1} \end{array} = \begin{array}{cc} (\cdot \uparrow_2) \star h & (\cdot \uparrow_2) \star g \end{array} \times \begin{array}{c} a_j \\ \hline d_j \end{array}$$

Analysis $\mathcal{W}_j$
Synthesis $\mathcal{W}_j^\top = \mathcal{W}_j^{-1}$